

CS 305 Lab Tutorial

Lab 5 DNS

Dept. Computer Science and Engineering
Southern University of Science and Technology

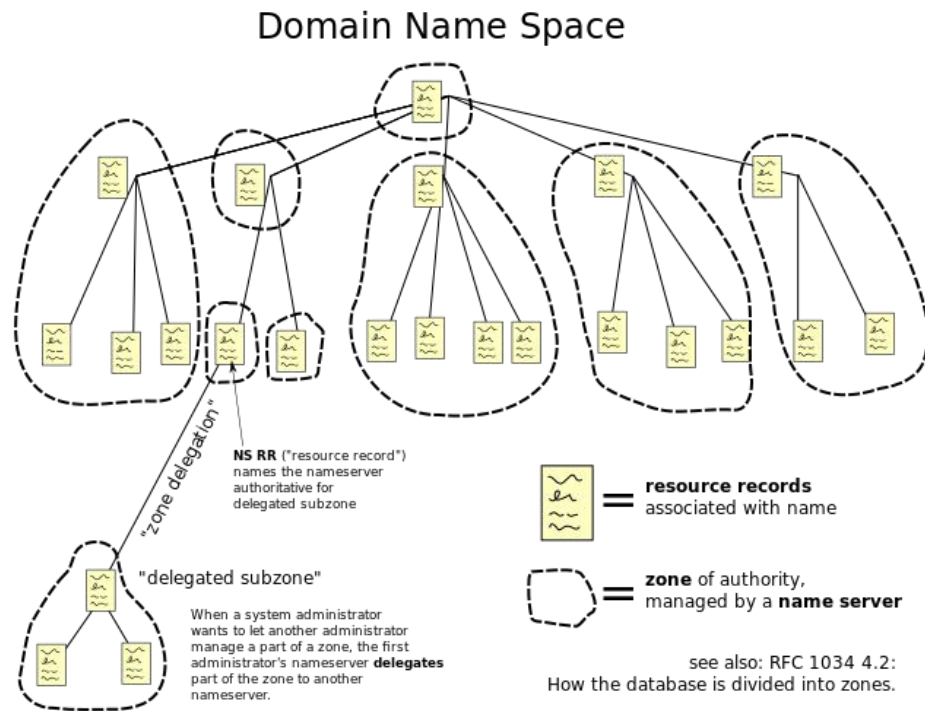
Topic

- DNS
 - DNS Message Structure
 - DNS Message head
 - RR in DNS
- EDNS (aka. Extension mechanisms for DNS)
 - DNSSEC
- Tool : dig

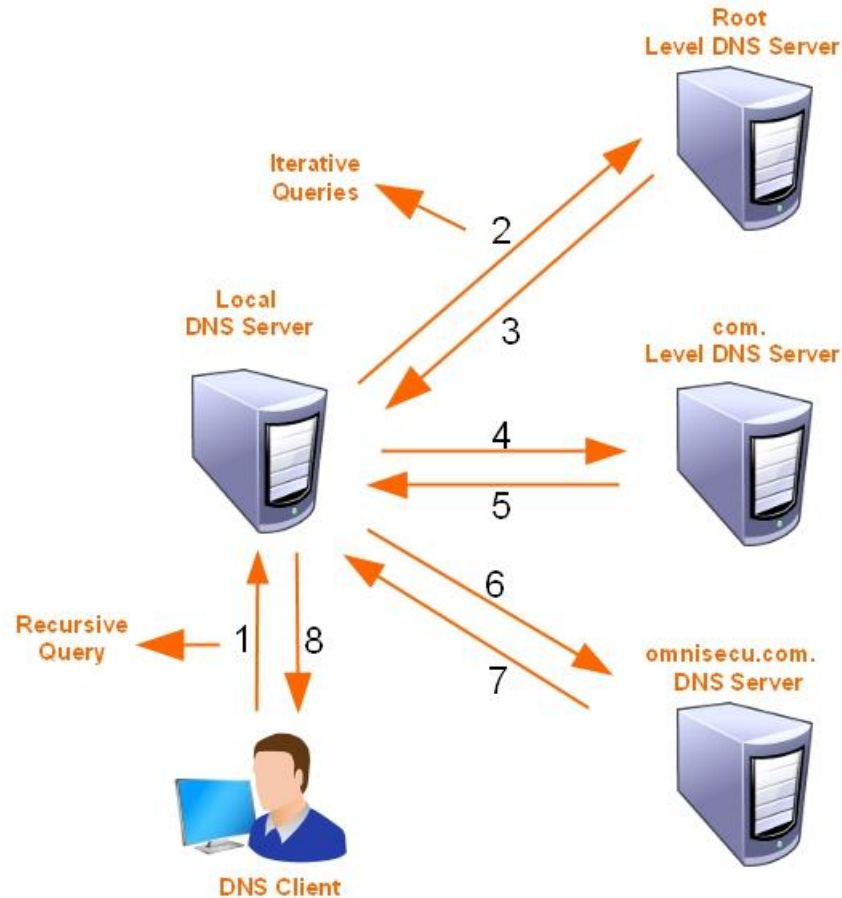
Part A.1

Domain Name System

- DNS is a distributed database.

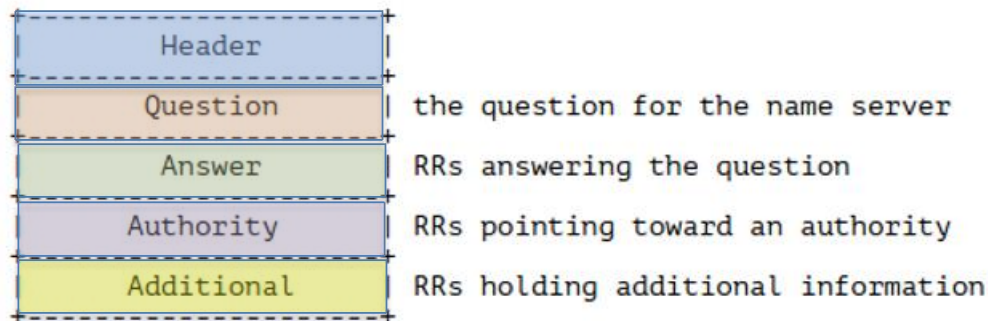


Recursive/Iterative Query



Part A.2

DNS Message Structure



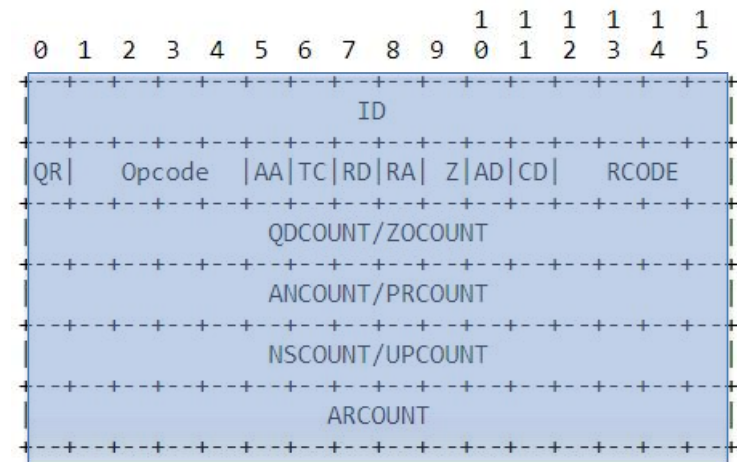
DNS协议报文格式

RFC 2929 DNS Message Headers

Domain Name System (DNS) IANA Considerations

- Set QR bit to 0 indicates the header is a query, otherwise is a response.
- OpCode 0 indicates this is a standard query.
- AA, TC, RD, RA, AD, CD stands for Authoritative Answer, Truncated, Recursion Desired, Recursion Available, Authentic Data, Checking Disabled.
- Z is a reserved flag.

OpCode	Name	Reference
0	Query	[RFC 1035]
1	IQuery (Inverse Query)	[RFC 1035]
2	Status	[RFC 1035]
3	available for assignment	
4	Notify	[RFC 1996]
5	Update	[RFC 2136]
6-15	available for assignment	

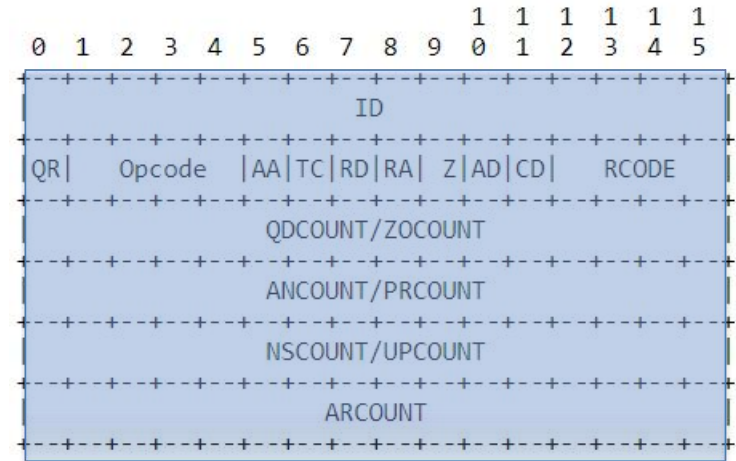


Example Structure Code in C:

```
struct DNS_HEADER { //DNS header structure
    unsigned short id; // identification number

    unsigned char qr :1; // query/response flag
    unsigned char opcode :4; // purpose of message
    unsigned char aa :1; // authoritative answer
    unsigned char tc :1; // truncated message
    unsigned char rd :1; // recursion desired
    unsigned char ra :1; // recursion available
    unsigned char z :1; // its z reserved
    unsigned char ad :1; // authenticated data
    unsigned char cd :1; // checking disabled
    unsigned char rcode :4; // response code

    unsigned short q_count; // number of question entries
    unsigned short ans_count; // number of answer entries
    unsigned short auth_count; // number of authority entries
    unsigned short add_count; // number of resource entries
};
```



Decode Message Header in Python

```
class DNSHeader:
```

```
    Struct = struct.Struct('!6H')
```

```
    def __init__(self):
```

```
        self.__dict__ = {
```

```
            field: None
```

```
            for field in ('ID', 'QR', 'OpCode', 'AA', 'TC', 'RD', 'RA', 'Z',  
                          'RCode', 'QDCount', 'ANCount', 'NSCount', 'ARCount')}]
```

```
    def parse_header(self, data):
```

```
        self.ID, misc, self.QDCount, self.ANcount, self.NScount, self.ARcount = DNSHeader.Struct.unpack_from(data)
```

```
        self.QR = (misc & 0x8000) != 0
```

```
        self.OpCode = (misc & 0x7800) >> 11
```

```
        self.AA = (misc & 0x0400) != 0
```

```
        self.TC = (misc & 0x200) != 0
```

```
        self.RD = (misc & 0x100) != 0
```

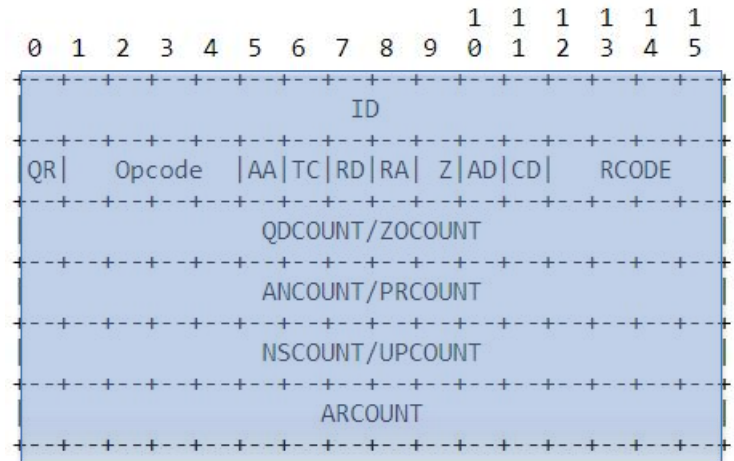
```
        self.RA = (misc & 0x80) != 0
```

```
        self.Z = (misc & 0x70) >> 4 # Never used
```

```
        self.RCode = misc & 0xF
```

```
    def __str__(self):
```

```
        return '<DNSHeader {}>'.format(str(self.__dict__))
```



A query message of DNS

nslookup www.baidu.com

dns.qry.name=="www.baidu.com"					
No.	Time	Source	Destination	Protocol	Info
83	7.091216	10.20.68.65	172.18.1.92	DNS	Standard query 0x0006
84	7.093045	172.18.1.92	10.20.68.65	DNS	Standard query response

> Frame 83: 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface 0
> Ethernet II, Src: IntelCor_5c:69:58 (90:61:ae:5c:69:58), Dst: JuniperN_ab:30:03 (40:71:83:ab:30:03)
> Internet Protocol Version 4, Src: 10.20.68.65 (10.20.68.65), Dst: 172.18.1.92 (172.18.1.92)
> User Datagram Protocol, Src Port: 55788 (55788), Dst Port: domain (53)

Domain Name System (query)

Transaction ID: 0x0006

Flags: 0x0100 Standard query

0... .. = Response: Message is a query
.000 0... .. = Opcode: Standard query (0)
.... ..0. = Truncated: Message is not truncated
.... ..1 = Recursion desired: Do query recursively
.... ..0.. = Z: reserved (0)
.... ..0 = Non-authenticated data: Unacceptable

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Just 1 question with no answer

Header

Queries

www.baidu.com: type A, class IN

Name: www.baidu.com

[Name Length: 13]

[Label Count: 3]

Type: A (Host Address) (1)

Class: IN (0x0001)

Question

No answer

No authority

No additional

[Response In: 84]

A response message of DNS

nslookup www.baidu.com

dns.qry.name=="www.baidu.com"

No.	Time	Source	Destination	Protocol	Info
84	7.093045	172.18.1.92	10.20.68.65	DNS	Standard query response 0x0006

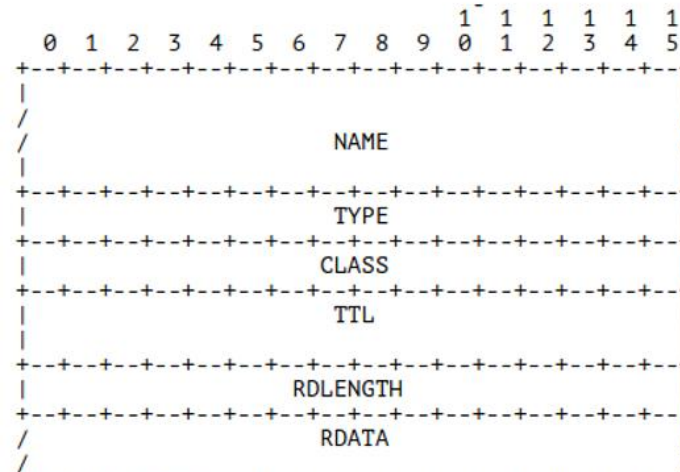
> Frame 84: 286 bytes on wire (2288 bits), 286 bytes captured (2288 bits) on interface 0
> Ethernet II, Src: JuniperN_ab:30:03 (40:71:83:ab:30:03), Dst: IntelCor_5c:69:58 (90:61:ae:5c:69:58)
> Internet Protocol Version 4, Src: 172.18.1.92 (172.18.1.92), Dst: 10.20.68.65 (10.20.68.65)
> User Datagram Protocol, Src Port: domain (53), Dst Port: 55788 (55788)
▼ Domain Name System (response)
Transaction ID: 0x0006
▼ Flags: 0x8180 Standard query response, No error
1... .. = Response: Message is a response
... .. = Opcode: Standard query (0)
... .. = Authoritative: Server is not an authority for domain
... .. = Truncated: Message is not truncated
... ..1... .. = Recursion desired: Do query recursively
... ..1... .. = Recursion available: Server can do recursive queries
... ..0... .. = Z: reserved (0)
... ..0... .. = Answer authenticated: Answer/authority portion was not authenticated by the
... ..0... .. = Non-authenticated data: Unacceptable
... ..0000 = Reply code: No error (0)
Questions: 1
Answer RRs: 3
Authority RRs: 5
Additional RRs: 4
> Queries
> Answers
> Authoritative nameservers
> Additional records
[Request in: 83]
[Time: 0.001829000 seconds]

Header

Question
Answer
Authority
Additional

Part A.3

RR in DNS



Resource record (RR) fields

Field	Description	Length (octets)
NAME	Name of the node to which this record pertains	Variable
TYPE	Type of RR in numeric form (e.g., 15 for MX RRs)	2
CLASS	Class code	2
TTL	Count of seconds that the RR stays valid (The maximum is $2^{31}-1$, which is about 68 years)	4
RDLENGTH	Length of RDATA field (specified in octets)	2
RDATA	Additional RR-specific data	Variable, as per RDLENGTH

RRs of Answers

nslookup www.baidu.com

No.	Time	Source	Destination	Protocol	Info
84	7.093045	172.18.1.92	10.20.68.65	DNS	Standard query response 0x0006
< >					
▼ Domain Name System (response)					
Transaction ID: 0x0006					
> Flags: 0x8180 Standard query response, No error					
Questions: 1					
Answer RRs: 3					
Authority RRs: 5					
Additional RRs: 4					
> Queries					
▼ Answers					
▼ www.baidu.com: type CNAME, class IN, cname www.a.shifen.com					
Name: www.baidu.com					
Type: CNAME (Canonical NAME for an alias) (5)					
Class: IN (0x0001)					
Time to live: 77					
Data length: 15					
CNAME: www.a.shifen.com					
▼ www.a.shifen.com: type A, class IN, addr 14.215.177.38					
Name: www.a.shifen.com					
Type: A (Host Address) (1)					
Class: IN (0x0001)					
Time to live: 168					
Data length: 4					
Address: www.a.shifen.com (14.215.177.38)					
▼ www.a.shifen.com: type A, class IN, addr 14.215.177.39					
Name: www.a.shifen.com					
Type: A (Host Address) (1)					
Class: IN (0x0001)					
Time to live: 168					
Data length: 4					
Address: www.a.shifen.com (14.215.177.39)					
> Authoritative nameservers					

all the answers share
the same structure:
name,type,class,ttl
and length

RRs of authoritative name servers

nslookup www.baidu.com

```
✓ Domain Name System (response)
  Transaction ID: 0x0006
  > Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 3
  Authority RRs: 5
  Additional RRs: 4
  > Queries
  > Answers
  ✓ Authoritative nameservers
    ✓ a.shifen.com: type NS, class IN, ns ns3.a.shifen.com
      Name: a.shifen.com
      Type: NS (authoritative Name Server) (2)
      Class: IN (0x0001)
      Time to live: 66
      Data length: 6
      Name Server: ns3.a.shifen.com
    > a.shifen.com: type NS, class IN, ns ns2.a.shifen.com
    > a.shifen.com: type NS, class IN, ns ns1.a.shifen.com
    > a.shifen.com: type NS, class IN, ns ns5.a.shifen.com
    > a.shifen.com: type NS, class IN, ns ns4.a.shifen.com
  > Additional records
  [Request In: 83]
  [Time: 0.001829000 seconds]
```

the value of rdata depend on
the type

RRs of Additional records

nslookup www.baidu.com

```
dns.gry.name=="www.baidu.com" Expression...
No.      Time      Source      Destination  Protocol Info
84 7.093045 172.18.1.92 10.20.68.65  DNS      Standard query response 0x0006

> Ethernet II, Src: JuniperN_ab:30:03 (40:71:83:ab:30:03), Dst: IntelCor_5c:69:58 (90:61:ae:5c:69:58)
> Internet Protocol Version 4, Src: 172.18.1.92 (172.18.1.92), Dst: 10.20.68.65 (10.20.68.65)
> User Datagram Protocol, Src Port: domain (53), Dst Port: 55788 (55788)
v Domain Name System (response)
  Transaction ID: 0x0006
  > Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 3
  Authority RRs: 5
  Additional RRs: 4
  > Queries
  > Answers
  > Authoritative nameservers
  v Additional records
    v ns1.a.shifen.com: type A, class IN, addr 61.135.165.224
      Name: ns1.a.shifen.com
      Type: A (Host Address) (1)
      Class: IN (0x0001)
      Time to live: 423
      Data length: 4
      Address: ns1.a.shifen.com (61.135.165.224)
    v ns2.a.shifen.com: type A, class IN, addr 220.181.33.32
      Name: ns2.a.shifen.com
      Type: A (Host Address) (1)
      Class: IN (0x0001)
      Time to live: 4
      Data length: 4
      Address: ns2.a.shifen.com (220.181.33.32)
  > ns3.a.shifen.com: type A, class IN, addr 112.80.255.253
  > ns4.a.shifen.com: type A, class IN, addr 14.215.177.229
[Request in: 83]
[Time: 0.001829000 seconds]
```

Part B

EDNS (aka. Extension mechanisms for DNS)

EDNS: a backward compatible mechanisms for allowing the DNS protocol to grow.

- The Domain Name System's wire protocol includes a number of **fixed** fields whose range has been or soon will be exhausted and does not allow clients to advertise their capabilities to servers.
- DNS (see [RFC1035]) specifies a Message Format and within such messages there are standard formats for encoding options, errors, and name compression. The maximum allowable size of a DNS Message is fixed.
- Many of DNS's protocol limits are too small for uses which are or which are desired to become common. There is no way for implementations to advertise their capabilities.

DNS 原始协议的问题 (RFC 1035)

作为 DNS 协议在 1983 年定义 (RFC 1035)，但有几个局限：

- 报文结构中很多字段的长度是固定的（例如字段最多 16 位）；
- UDP 报文大小最多只能是 512 字节；
- 没有机制让客户端告诉服务器“我支持哪些新特性”；
- 无法传输新的选项、错误码或压缩机制。

这些问题在最初的网络环境中没关系，但现在：

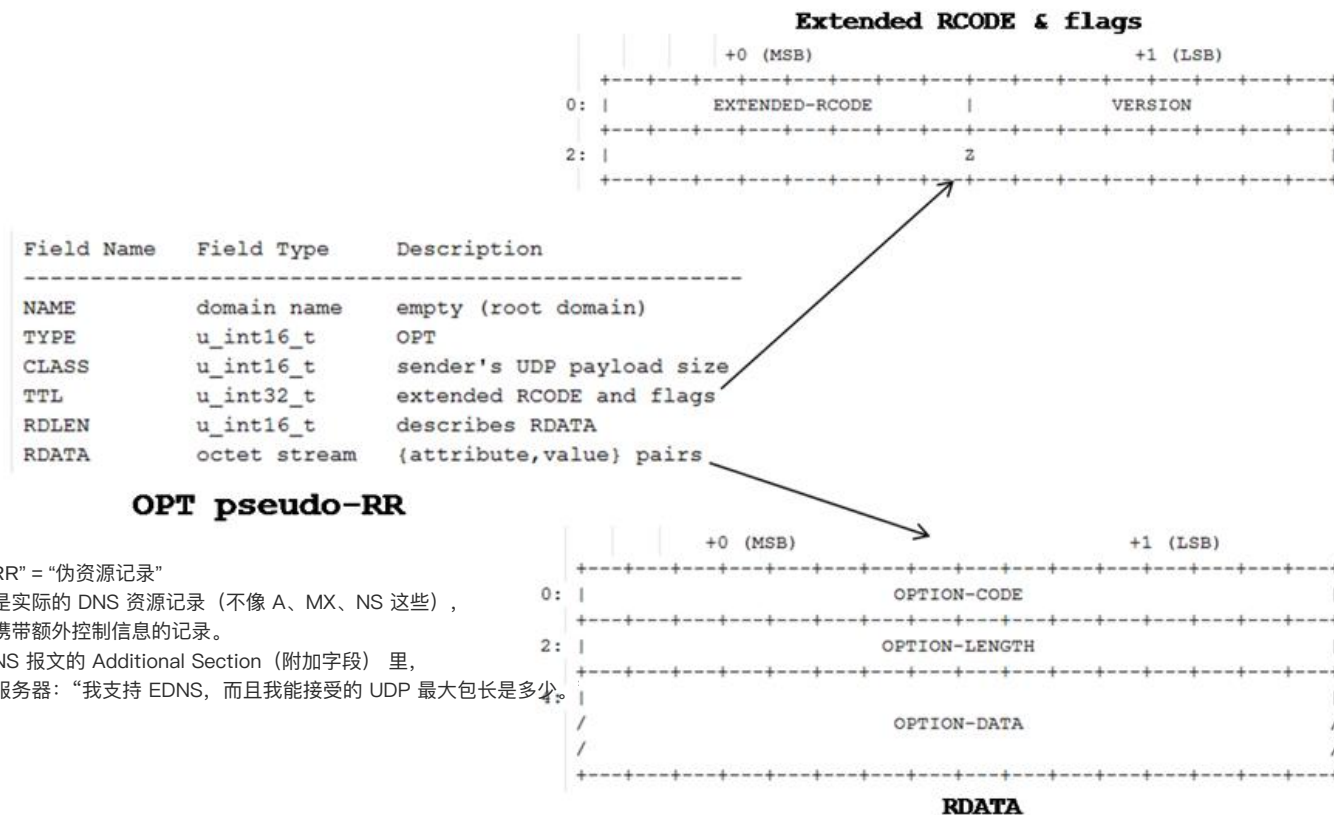
- DNS 查询数据越来越大（例如带签名的 DNSSEC、IPv6 地址等）；
- 更多的新功能（如安全、加密、QoS）需要更大的灵活性；
- 所以原始 DNS 格式太小、太死板了。

<https://tools.ietf.org/html/rfc2671>

EDNS

EDNS 的核心思想是：
在保持向后兼容的前提下，扩展 DNS 协议的能力。
也就是说：
旧的 DNS 服务器仍能理解 EDNS 报文（不会崩溃）；
新的服务器可以“互相识别”出对方支持扩展功能；
并且可以交换更多信息（例如更大的 UDP 包、DNSSEC 选项等）。

One **OPT pseudo-RR** can be added to the **additional data section** of either a request or a response. An OPT is called a pseudo-RR because it pertains to a particular transport level message and not to any actual DNS data.



EDNS query

Domain Name System (query)

Transaction ID: 0xe9d8

> Flags: 0x0120 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 1

> Queries

Additional records

<Root>: type OPT

Name: <Root>

Type: OPT (41)

UDP payload size: 4096

Higher bits in extended RCODE: 0x00

EDNS0 version: 0

Z: 0x0000

0... = DO bit: Cannot handle DNSSEC security RRs

.000 0000 0000 0000 = Reserved: 0x0000

Data length: 12

> Option: COOKIE

Field Name	Field Type	Description
NAME	domain name	empty (root domain)
TYPE	u_int16_t	OPT
CLASS	u_int16_t	sender's UDP payload size
TTL	u_int32_t	extended RCODE and flags
RDLEN	u_int16_t	describes RDATA
RDATA	octet stream	{attribute,value} pairs

EDNS response

Domain Name System (response)

Transaction ID: 0xe9d8

> Flags: 0x8180 Standard query response, No error

Questions: 1

Answer RRs: 3

Authority RRs: 5

Additional RRs: 5

> Queries

> Answers

> Authoritative nameservers

Additional records

> ns1.a.shifen.com: type A, class IN, addr 61.135.165.224

> ns2.a.shifen.com: type A, class IN, addr 220.181.33.32

> ns3.a.shifen.com: type A, class IN, addr 112.80.255.253

> ns5.a.shifen.com: type A, class IN, addr 180.76.76.95

<Root>: type OPT

Name: <Root>

放置位置

Type: OPT (41)

DNS 报文的 Additional Section

UDP payload size: 4096

Higher bits in extended RCODE: 0x00

EDNS0 version: 0

Z: 0x0000

0... .. = DO bit: Cannot handle DNSSEC security RRs

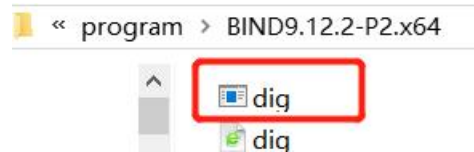
.000 0000 0000 0000 = Reserved: 0x0000

Data length: 0

Field Name	Field Type	Description
NAME	domain name	empty (root domain)
TYPE	u_int16_t	OPT
CLASS	u_int16_t	sender's UDP payload size
TTL	u_int32_t	extended RCODE and flags
RDLEN	u_int16_t	describes RDATA
RDATA	octet stream	{attribute,value} pairs

Part C dig

- **dig** is a flexible tool for interrogating DNS name servers.
 - It performs DNS lookups and displays the answers that are returned from the name server(s) that were queried.
 - Most DNS administrators use **dig** to troubleshoot DNS problems because of its flexibility, ease of use and clarity of output.



Bind is a Toolset which includes dig as a component
Bind could be get from <https://www.isc.org/bind/>

Using dig

参数	含义	📄
@server	要查询的 DNS 服务器地址 (如 8.8.8.8)。如果不写, 默认使用本机配置的 DNS。	
name	你想要查询的域名 (如 www.baidu.com)。	➔
type	查询类型, 例如 A、MX、CNAME、NS、ANY 等。默认是 A 记录。	

- A typical invocation of dig looks like: **dig @server name type**
 - **server**: the name or IP address of the name server to query. This can be an IPv4 address or an IPv6 address. When the supplied server argument is a hostname, dig resolves that name before querying that name server.
 - **name**: the name of the resource record that is to be looked up.
 - **Type** indicates what type of query is required: ANY, A, MX, SIG, etc. and A record is default when using dig.
 - Some useful options: -h, +tcp, +noedns, +bufsize, +trace, .etc.

```
D:\>dig @ns2.sustech.edu.cn www.baidu.com a +short
www.a.shifen.com.
163.177.151.109
163.177.151.110

D:\>dig @ns2.sustech.edu.cn www.baidu.com cname +short
www.a.shifen.com.

D:\>dig @ns2.sustech.edu.cn www.baidu.com mx +short
www.a.shifen.com.
```

Tips 1

- Use nslookup command to check your local DNS server.

```
C:\Users\wqhrb>nslookup www.baidu.com
服务器:  ns1.sustech.edu.cn
Address:  172.18.1.92

非权威应答:
名称:     www.a.shifen.com
Addresses: 14.215.177.38
           14.119.104.189
Aliases:  www.baidu.com
```

- Some common public DNS server
 - 114 DNS Preferred address: 114.114.114.114, alternative address: 114.114.115.115
 - Baidu DNS IPv4 address: 180.76.76.76, IPv6 address: 2400:da00::6666
 - Google Public DNS IPv4 address: 8.8.8.8, DNS IPv6: 2001:4860:4860::8888

Practice 5.1

- Make the query of “www.sina.com.cn” by using “dig” with option “+trace” .
- Capture screenshots on the command and its output, answer the questions by analyzing the packets:
 - How many queries are sent from the local host? Do they share the same “transaction id”?
 - How many responses are received by local host?
 - what's the value of ‘RD’ field in the query? what's the value of ‘RA’ field in responses from the local DNS server?
 - which server sent the last response, is it the local DNS server of the local host or is it the Authoritative DNS server?
 - List the name, IP address and port number of the server which sent the last response to the local host.
 - Is there any answer in the response? What's the value of ‘AA’ field in this response ?
 - try the same query again
 - At this time, is the last response from the same server as the last response in the previous query? If they are different, what is the reason for this? will it bring any benefits?

QR	Opcode	AA	TC	RD	RA	Z	AD	CD	RCODE
----	--------	----	----	----	----	---	----	----	-------

Tips 2

- Use “ipconfig /displaydns” to list DNS buffer of local host.

```
C:\Users\wq>ipconfig /displaydns
```

Windows IP 配置

ping.pinyin.sogou.com

```
-----
记录名称. . . . . : ping.pinyin.sogou.com
记录类型. . . . . : 5
生存时间. . . . . : 36
数据长度. . . . . : 8
部分. . . . . : 答案
CNAME 记录 . . . . . : ping.sogou.com
```

sakai.sustech.edu.cn

```
-----
记录名称. . . . . : sakai.sustech.edu.cn
记录类型. . . . . : 1
生存时间. . . . . : 89
数据长度. . . . . : 4
部分. . . . . : 答案
A (主机)记录 . . . . : 116.7.234.94
```

- Use “ipconfig /flushdns” to clear DNS buffer.

```
C:\Users\wq>ipconfig /flushdns
```

Windows IP 配置

已成功刷新 DNS 解析缓存。

Practice 5.2(optional)

- Make an DNS query for “www.bilibili.com” with 'EDNS0' option.
- Capture the packages using Wireshark and answer the following questions:
 - What is the content of the query message
 - what's the destination IP address and destination port of the query?
 - what's the name, type and class of this query?
 - what's the opcode of these query, what does it mean?
 - Is there any additional RR, what's the type of the RR?
 - What is the content of the response message
 - Is there any answers, what's life time of each answer?
 - Is there any authority RRs, what's the type of each RR?

Practice 5.3

Implement a DNS client

- Function:
 - Invoke DNS queries.
 - Support common query types: A, AAAA, CNAME, NS, MX
 - EDNS implementation is not required.
 - Implement both queries with 'RD' equals to 0 and 1.
 - Check out the response.
 - Display the answer, check who sends the answer? Whether the answer is from authority Name Server or not? How to judge that?
 - Do the iterative query while 'RD' equals to 0 to get the desired answer which is identified in the query.
- Tips:
 - using “dig” with “+trace” option to invoke the iterative query, using Wireshark to capture the packets of the iterative query.
 - while using “dig” with “+trace” option, the desired answer may not be got. for example: the desired answer is A type record in query while “dig” ends the iterative query while it got the 1st answer which maybe the CNAME type.

Tips 3: Using dns.resolver of python(1)

Using pip to install dnspython

- pip is the package installer for Python. You can use pip to install packages from the Python Package index and other indexes.

```
C:\Users\wiri>pip install dnspython
Collecting dnspython
  Downloading https://files.pythonhosted.org/packages/a6/72/209e18bdfedfd78c6994e9ec96981624a5ad7738524dd474237268422ct
/dnspython-1.15.0-py2.py3-none-any.whl (177kB)
  100% |#####| 184kB 18kB/s
Installing collected packages: dnspython
Successfully installed dnspython-1.15.0
```

A demo of using query of dns.resolver

If 'pip' is not installed on your computer, get it from
<https://pypi.org/project/pip/>

Get more information about dnspython, get it from
<https://pypi.org/project/dnspython/>

```
>>> import dns.resolver
>>> dns.resolver.query("www.baidu.com", 'a')
<dns.resolver.Answer object at 0x000002316AF22860>
>>> a = dns.resolver.query("www.baidu.com", 'a')
>>> a
<dns.resolver.Answer object at 0x000002316AF277F0>
>>> for i in a.response.answer:
...     for j in i.items:
...         print(j)
...
www.a.shifen.com.
163.177.151.110
163.177.151.109
>>>
```

Tips 3: Using dns.resolver of python(2)

query in dns.resolver of python

- `query(self, qname, rdtype=1, rdclass=1, tcp=False, source=None, raise_on_no_answer=True, source_port=0)`
 - Query nameservers to find the answer to the question.
 - The `qname`, `rdtype`, and `rdclass` parameters may be objects of the appropriate type, or strings that can be converted into objects of the appropriate type. E.g. For `rdtype` the integer 2 and the string 'NS' both mean to query for records with DNS rdata type NS.
- Parameters:
 - `qname` (`dns.name.Name` object or string) - the query name
 - `rdtype` (int or string) - the query type
 - `rdclass` (int or string) - the query class
 - `tcp` (bool) - use TCP to make the query (default is False).
 - `source` (IP address in dotted quad notation) - bind to this IP address (defaults to machine default IP).
 - `raise_on_no_answer` (bool) - raise `NoAnswer` if there's no answer (defaults is True).
 - `source_port` (int) - The port from which to send the message. The default is 0.

Tips 4: UDP socket programming

UDP Server

```
udp_c.py x udp_s.py x
1  from socket import *
2  serverPort = 12000
3  serverSocket = socket(AF_INET, SOCK_DGRAM)
4  serverSocket.bind(('', serverPort))
5  print ("The server is ready to receive")
6  while True:
7      message, clientAddress = serverSocket.recvfrom(2048)
8      modifiedMessage = message.decode().upper()
9      serverSocket.sendto(modifiedMessage.encode(), clientAddress)
```

```
d:\python_test>python udp_s.py
The server is ready to receive
```

UDP Client

```
udp_c.py x udp_s.py x
1  from socket import *
2  serverName = '127.0.0.1'
3  serverPort = 12000
4  clientSocket = socket(AF_INET, SOCK_DGRAM)
5  message = input('Input lowercase sentence:')
6  clientSocket.sendto(message.encode(), (serverName, serverPort))
7  modifiedMessage, serverAddress = clientSocket.recvfrom(2048)
8  print(modifiedMessage.decode())
9  clientSocket.close()
```

```
d:\python_test>python udp_c.py
Input lowercase sentence:azs
AZS
```